Example 6.38 If the equation $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines, find (i) the value of λ and the separate equations of the lines (ii) point of intersection of the lines (iii) angle between the lines

Solution:

(i) general equation is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

given equation is $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$

Comparing the given equation with the general equation of the second degree we have

$$a = \lambda, b = 12, c = -3, h = -5, g = \frac{5}{2}, f = -8$$

Now applying the condition for pair of straight lines

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\text{i.e. } \lambda(12)(-3) + 2(-8)\left(\frac{5}{2}\right)(-5) - \lambda(-8)^2 - 12\left(\frac{5}{2}\right)^2 - (-3)(-5)^2 = 0$$

$$\Rightarrow -36\lambda + 200 - 64\lambda - 75 + 75 = 0, \quad \Rightarrow \lambda = 2$$

$\therefore$ The equation is $2x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$.
Let us first factorize the terms of second degree terms from the above equation,

$$\text{we get} \qquad 2x^2 - 10xy + 12y^2 \equiv (x - 2y)(2x - 6y)$$

$$\therefore \; 2x^2 - 10xy + 12y^2 + 5x - 16y - 3 \equiv (x - 2y + c_1)(2x - 6y + c_2)$$

Equating like terms, we get

$$2c_1 + c_2 = 5, \quad 3c_1 + c_2 = 8, \quad c_1c_2 = -3$$

Solving first two equation, we get $c_1 = 3, c_2 = -1$.

$\therefore$ The separate equations of the lines are

$$x - 2y + 3 = 0 \quad \text{and} \quad 2x - 6y - 1 = 0$$

(ii) Point of intersection of the lines is given by solving the two equation of the lines, we get

$$(x, y) = \left(-10, -\frac{7}{2}\right) \left(\text{or use the formula } \left(\frac{hf - bg}{ab - h^2}, \frac{hg - af}{ab - h^2}\right)\right)$$

(iii) Angle between the lines is given by

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$$

$$= \frac{2\sqrt{25 - 24}}{2 + 12} = \frac{1}{7}$$

$$\therefore \; \theta = \tan^{-1}\left(\frac{1}{7}\right)$$